Lumbar spine segmentation: training on VerSe instead of TotalSegmentator

1 Setup

The pipeline, network (residual-encoder U-Net), loss, optimiser, 11 000 training steps, checkpoint rule and test sets are those of the original project. Only the training data change, and with them the voxel size:

- **TotalSegmentator-trained** (original project): 902 TotalSegmentator v3 training scans, 1.5 mm voxels (the native TotalSegmentator size), batch 3.
- **VerSe-trained** (this project): the VerSe19 training set, 80 scans of 67 patients; 62 usable after excluding 15 scans without a lumbar vertebra and 3 with an L6 (the network has no L6 class); 56 for training and 6 for choosing the checkpoint, split by patient. Voxel size **1 mm**: 39 of the 80 scans are already 1 mm isotropic, whereas at 1.5 or 2 mm all 80 would be resampled. One 1 mm patch (192×192×320 voxels, the same 19.2×19.2×32 cm) fills the 24 GB GPU, so batch 1.

Table 1: Anomalies found by the automatic check (Table 1 of the original report, same rules), with the detection and handling of each. Counts are numbers of scans: TS = TotalSegmentator (1939), VS = VerSe19 validation + test (80), **VS train = VerSe19 training set (80), used here for training**. Upper block: header; lower block: image and label content. [a]Five VS-train scans have exactly 3× (1 × 1 × 3 mm) and are not counted. [b]The network has no L6 class; the 3 VS-train scans with L6 are not used for training.

Anomaly	How detected	How handled	TS	VS	**VS train**
Image not stored axially (PIR, PSR, LAS)	orientation codes of the affine	resampled to RAS	0	48	**41**
Thick slices (more than 3× the in-plane size)[a]	voxel sizes	smoothed before downsampling	0	4	**0**
Only qform or only sform set	header codes	the one that is set is used	1939	80	**80**
CT stored as float, HU not whole numbers	data type and values	accepted (resampled by provider)	0	77	**73**
Label and CT affines differ by ≤ 0.001 mm	affine difference	treated as the same grid	228	36	**28**
Label mask stored as float	data type	rounded to whole numbers	0	1	**0**
Two masks claim the same voxel (mostly S1 and L5)	overlap of the masks	the lumbar level wins	817	0	**0**
Padding below −2000 HU outside the scan circle	share of such voxels	set to air by the HU window	104	0	**0**
Metal near the spine	> 0.5 ml above 3000 HU	clipped at 1500 HU, flagged	124	4	**4**
Vertebra cut off by the scan edge	mask touches the border	kept; Dice also reported without	431	8	**4**
No lumbar vertebra in the scan	label content	excluded from training	720	13	**15**
Sixth lumbar vertebra (L6)	VerSe label 25	own class; excluded from training[b]	0	3	**3**

Table 2: Dice per vertebra on the two test sets, mean / median (Table 2 of the original report, with a row for each model). *Limit*: Dice of the ground truth after resampling to the model's voxel size and back, i.e. the highest score possible at that voxel size. *n*: number of scored vertebrae.

Test set	Trained on (voxel size)	L1	L2	L3	L4	L5	All	Limit	*n*
TotalSegmentator test (78 scans)	TotalSegmentator (902 scans) (1.5 mm)	.88 / .97	.90 / .97	.92 / .97	.91 / .97	.87 / .96	**.89 / .97**	1.00	328
	VerSe19 training set (56 scans) (1 mm)	*not evaluated yet*							
VerSe19 val + test (67 scans)	TotalSegmentator (902 scans) (1.5 mm)	.90 / .94	.91 / .94	.92 / .94	.91 / .94	.89 / .93	**.91 / .94**	.96	305
	VerSe19 training set (56 scans) (1 mm)	.58 / .77	.72 / .94	.83 / .94	.83 / .94	.86 / .93	**.76 / .93**	1.00	305

Table 3: Further comparison numbers. *Counted one level off*: renaming all predicted levels by one raises the scan's mean Dice by more than 0.1. *Whole-vertebra false positives*: predicted levels absent from the ground truth and at least 20 ml large. *Body centre*: distance between the predicted and the ground-truth vertebral body centre, median / 90th percentile.

	TotalSegmentator test (78 scans)		VerSe19 val + test (67 scans)	
Trained on	TotalSegmentator	VerSe19 train	TotalSegmentator	VerSe19 train
Dice, complete vertebrae only (mean)	.92	–	.91	.77
Scans counted one level off	2	–	1	7
Scans with a false-positive level	52	–	23	23
Whole-vertebra false positives	11	–	26	72
Body centre distance (mm)	0.3 / 15.7	–	0.6 / 3.9	0.9 / 34.3

2 Findings

- **On a typical VerSe scan both models segment equally well** (median Dice .93 vs .94), although the VerSe-trained model saw only 56 training scans; at 1 mm VerSe also loses nothing to resampling (limit 1.00 vs .96).
- **On average the VerSe-trained model is clearly worse** (mean Dice .76 vs .91). The loss is in *naming*, not in outlining: 7 scans are counted one level off (TotalSegmentator-trained: 1), and L1 suffers most (mean .58 vs .90).
- **It labels neck and upper thoracic vertebrae as lumbar far more often**: 72 whole-vertebra false positives against 26. Underneath them in the ground truth lie neck vertebrae C1–C7 in 32 cases (TotalSegmentator-trained: 8), thoracic vertebrae in 13 (5), unlabelled bone in 21 (12) and real lumbar vertebrae with the wrong level name in 6 (1). Many VerSe test scans show the whole spine. Both models were trained on lumbar crops only (T12 to sacrum), but with 56 instead of 902 scans the VerSe-trained model has learned far less well what is *not* a lumbar vertebra. As the clean-up keeps only the largest piece per class, a wrongly named neck vertebra can even replace the real L1.
- **Body centres**: the median distance is similar (0.9 vs 0.6 mm), but the 90th percentile is 34 mm against 4 mm: the same naming errors put a "body centre" on the wrong vertebra.
- **Takeaway**: here, many training scans from another source beat few scans from the target source. 56 scans are not enough to learn level counting on whole-spine scans; more scans, training patches outside the lumbar crop, or a rule-based level check would be the next steps.

The TotalSegmentator test set is not yet evaluated for the VerSe-trained model (60 of 109 scans done); its rows are filled in automatically once the evaluation is complete.